

1) Reinforcement learning is a type of machine learning in which an agent learns by with an environment and receiving rewards and Penalties of the actions.

→ To learn optimal Policy that maximizes the Cumulative long-term reward.

2) A Markov decision Process is a mathematical model used for sequential Process where the next state depends only on current state and action.

Components:

- States
- Actions
- Transition Probability
- Reward function
- Discount factor

3) Bellman Equation for state value function:

$$V^{\pi}(s) = \sum_a \pi(a|s) \sum_{s'} P(s'|s,a) [R(s,a,s') + \gamma V^{\pi}(s')]$$

4) Exploration:

- Tries New actions to gain knowledge
- Focuses on learning.

Exploitation:

- Chooses the best known action.

→ Focuses on maximizing reward.

5) The Multi-armed Bandit Problem is an RL Problem where an agent chooses among multiple actions (arms) with unknown rewards to maximize total reward by balancing exploration and exploitation.

6) Dynamic Programming is a method for solving MDPs by breaking problems into smaller subproblems using Bellman's equation.

→ Policy Evaluation

→ Policy Iteration

→ Value Iteration.

7) Temporal difference learning is model-free RL method that updates value estimates using the current reward and the estimated value of next state.

$$V(s_t) \leftarrow V(s_t) + \alpha [R_{t+1} + \gamma V(s_{t+1}) - V(s_t)]$$

8) SARSA:

→ On Policy algorithm

→ Updates using the action actually taken.

→ More Conservative and safer.

Deep Q-Network is a reinforcement learning algorithm that uses a deep Neural network to approximate the Q-value function instead of storing Q-values in a table.

Advantages Over tabular Q-learning:

- Can Handle large or Continuous state spaces.
- Generalizes to unseen states
- More scalable and efficient for complex Problems.

10) Double Deep Q-Network is an Improved version of DQN that uses two networks to estimate Q-values more Accurately.

Why it is Preferred Over DQN

- Reduces Q-value overestimation
- Provides more stable and accurate learning
- Improves Overall performance in many RL tasks.